Iteration 5

CIS 320

Team: Card Script

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Table of Contents

1. Class Diagram *3*

1.2 Class Diagram Narrative **3**

1.3 Class Diagram **4**

2. Database Design *5*

2.1 Database Diagram **5**

2.2 Database Narrative **6**

2.3 Database Definitions **6**

3. User Interface Design *7*

3.1 User Interface Navigation Diagram **7**

3.2 User Interface Diagram Narrative **8**

3.3 Screen Layouts **9**

3.3.A Data Capture Forms 9

3.3.B Data Presentation Forms 10

4. Gantt Chart *11*

4.1 Gantt Chart Diagram **11**

4.2 Gantt Chart Narrative **12**

5. User Interface Prototypes *13*

1. Class Diagram
1.2 Class Diagram Narrative

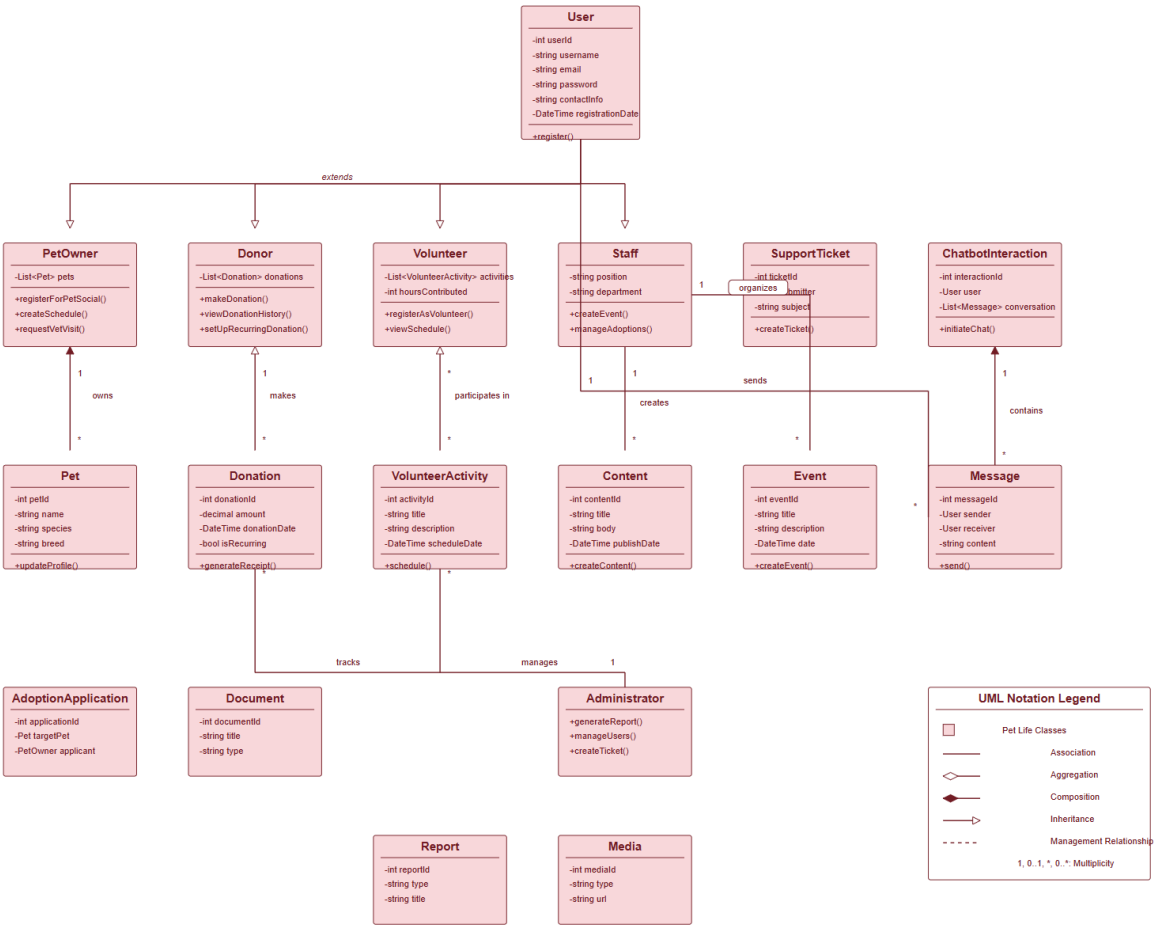
The Pet4Life platform is designed as a comprehensive solution for pet welfare, adoption, and community support. At the core of the system is the User class, which stores essential information like user identification, authentication credentials, and contact details. This central class branches into specialized user roles including PetOwner, Donor, Volunteer, and Staff, each extending the base User functionality with role-specific capabilities.

PetOwners interact primarily with their Pet records, which contain details about animals such as breed, species, and health information. They can also initiate AdoptionApplications when interested in adding new pets to their families. These applications are associated with relevant Documents that support the adoption process. Donors, meanwhile, create Donations that track financial contributions to the organization, maintaining details about amounts, dates, and specific program designations.

The Volunteer role facilitates engagement with VolunteerActivities, implementing a many-to-many relationship that allows volunteers to participate in multiple activities while each activity can accommodate numerous volunteers. Staff members serve as content creators and event organizers, authoring Content entries and managing Events for the community. Some staff members have extended capabilities through the Administrator role, which they inherit from Staff and adds reporting and system management functions. Administrators generate Reports that provide insights into platform operations and outcomes.

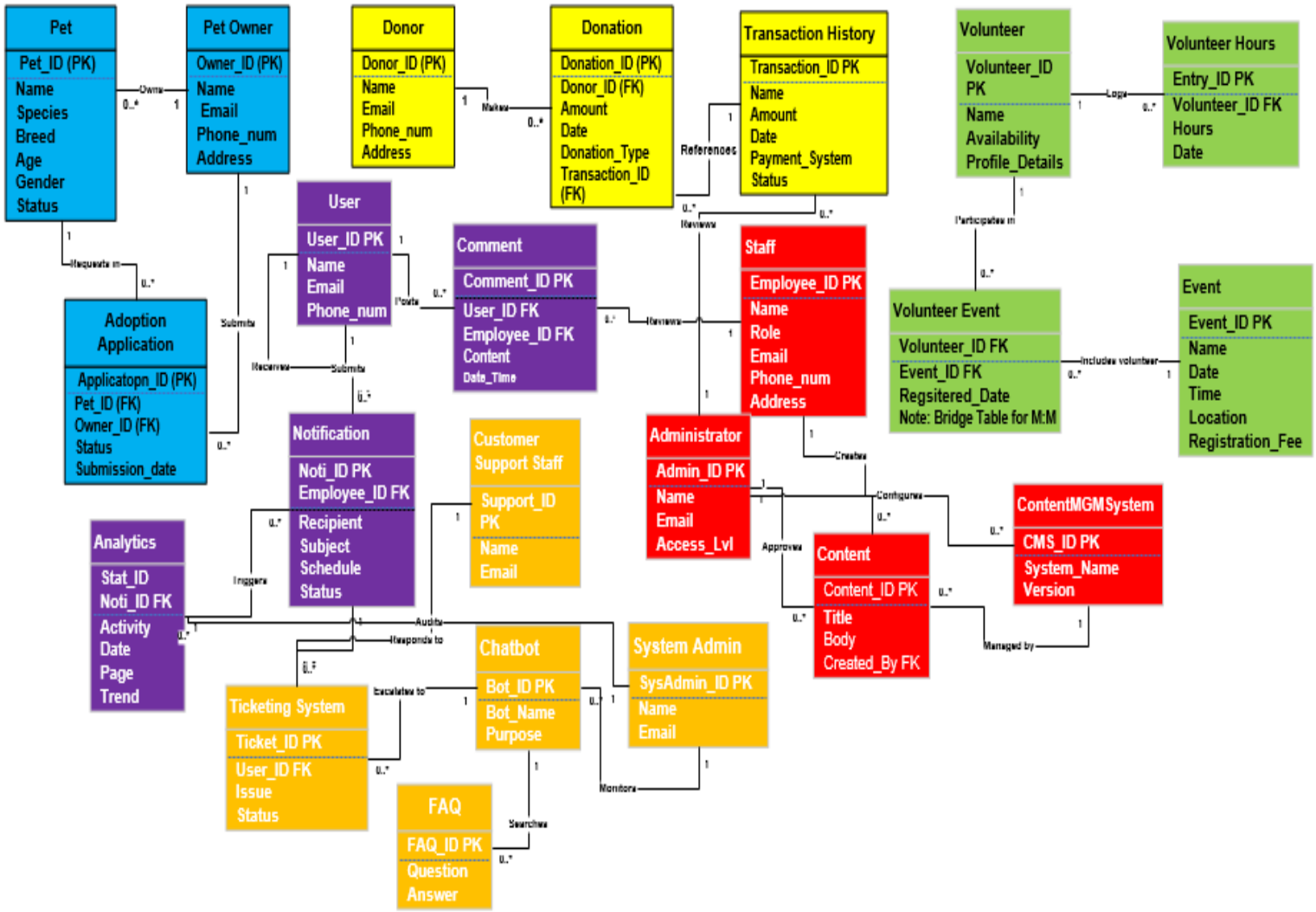
Supporting the user experience are SupportTickets and ChatbotInteractions, which provide mechanisms for users to receive assistance with platform functions. Messages facilitate communication between users, while Media elements enhance Content with visual and audio resources. This well-structured architecture enables Pet4Life to deliver a seamless experience for pet adoption, welfare support, volunteering, and community engagement, with clear relationship pathways between users, activities, and supporting data.

1.3 Class Diagram



2. Database Design

2.1 Database Diagram



2.2 Database Narrative

The database was designed to support the full operational workflow of Pets4Life, with a focus on pet adoption, donation management, volunteer tracking, content publishing, user engagement, and automated support services. The adoption process is driven by the connection between PetOwner, Pet, and AdoptionApplication, solving a real issue where owners needed to register multiple pets for adoption or appointments. Financial contributions are tracked through the Donor, Donation, and TransactionHistory tables, with administrative oversight added by linking the Administrator entity to reviewed transactions.

Volunteer activity is handled through Volunteer, VolunteerHours, Event, and the bridge table VolunteerEvent, which manages the many-to-many relationship between volunteers and events. User interaction is captured through the User, Comment, Notification, and Analytics entities, allowing Pets4Life to monitor engagement and platform responsiveness. Internally, Staff create content, Administrator roles approve it, and the ContentManagementSystem manages publishing. Additional automation and support needs are addressed through Chatbot, FAQ, TicketingSystem, Customer Support Staff, and System Admin, which together form a responsive help system and technical infrastructure.

Every entity in the database is directly connected to another, ensuring full normalization and data integrity. This structure addresses the organization’s real-world pain points—such as donation transparency, scalable volunteer tracking, and multi-pet registration—while remaining scalable, secure, and easy to manage moving forward.

2.3 Database Definitions

Pets

Attribute	Definition	Type	Size	Key
Pet_ID	Unique ID for the pet	Int	4 Bytes	Primary
Name	Pet's name	String	255 Bytes	No
Species	Species of the pet	String	255 Bytes	No
Breed	Breed of the pet	String	255 Bytes	No
Age	Age of the pet	Int	4 Bytes	No
Gender	Gender of the pet	String	10 Bytes	No
Status	Adoption status	String	255 Bytes	No

Pet Owner

Attribute	Definition	Type	Size	Key
Owner_ID	Unique ID for the pet owner	Int	4 Bytes	Primary
Name	Owner's name	String	255 Bytes	No
Email	Owner's email	String	255 Bytes	No
Phone_num	Owner's phone number	String	15 Bytes	No
Address	Owner's address	String	255 Bytes	No

Adoption Application

Attribute	Definition	Type	Size	Key
Application_ID	Unique application ID	Int	4 Bytes	Primary
Pet_ID	ID of the pet being adopted	Int	4 Bytes	Foreign
Owner_ID	ID of the applicant (pet owner)	Int	4 Bytes	Foreign
Status	Application status	String	50 Bytes	No
Submission_date	Date of application submission	Date	8 Bytes	No

User

Attribute	Definition	Type	Size	Key
User_ID	Unique user ID	Int	4 Bytes	Primary

Name	User's name	String	255 Bytes	No
Email	User's email	String	255 Bytes	No
Phone_num	User's phone number	String	15 Bytes	No

Analytics

Attribute	Definition	Type	Size	Key
Stat_ID	Unique stat ID	Int	4 Bytes	Primary
Noti_ID	Notification reference	Int	4 Bytes	Foreign
Activity	Tracked activity	String	255 Bytes	No
Date	Activity date	Date	8 Bytes	No
Page	Page visited	String	255 Bytes	No
Trend	Observed trend	String	255 Bytes	No

Notification

Attribute	Definition	Type	Size	Key
Noti_ID	Notification ID	Int	4 Bytes	Primary
Employee_ID	Staff reference	Int	4 Bytes	Foreign
Recipient	Recipient's name/email	String	255 Bytes	No
Subject	Subject line	String	255 Bytes	No
Schedule	Scheduled time	Date	8 Bytes	No
Status	Notification status	String	50 Bytes	No

Comment

Attribute	Definition	Type	Size	Key
Comment_ID	Unique comment ID	Int	4 Bytes	Primary
User_ID	ID of commenting user	Int	4 Bytes	Foreign
Employee_ID	Staff reference	Int	4 Bytes	Foreign
Content	Comment content	Text	Unlimited	No
Date_Time	Date and time of comment	DateTime	8 Bytes	No

Staff

Attribute	Definition	Type	Size	Key
Employee_ID	Unique staff ID	Int	4 Bytes	Primary
Name	Staff member's name	String	255 Bytes	No
Role	Job role/title	String	100 Bytes	No
Email	Staff email	String	255 Bytes	No
Phone_num	Phone number	String	15 Bytes	No
Address	Address	String	255 Bytes	No

Donor

Attribute	Definition	Type	Size	Key
Donor_ID	Unique donor ID	Int	4 Bytes	Primary
Name	Donor name	String	255 Bytes	No
Email	Donor email	String	255 Bytes	No
Phone_num	Donor phone number	String	15 Bytes	No
Address	Donor address	String	255 Bytes	No

Donation

Attribute	Definition	Type	Size	Key
Donation_ID	Unique donation ID	Int	4 Bytes	Primary
Donor_ID	Donor reference	Int	4 Bytes	Foreign
Amount	Donated amount	Float	4 Bytes	No
Date	Donation date	Date	8 Bytes	No
Donation_Type	Type (cash, item, etc.)	String	100 Bytes	No
Transaction_ID	Reference to transaction	Int	4 Bytes	Foreign

Transaction History

Attribute	Definition	Type	Size	Key
Transaction_ID	Unique transaction ID	Int	4 Bytes	Primary
Name	Donor name	String	255 Bytes	No
Amount	Transaction amount	Float	4 Bytes	No
Date	Transaction date	Date	8 Bytes	No
Payment_System	Payment method used	String	100 Bytes	No
Status	Payment status	String	50 Bytes	No

Volunteer

Attribute	Definition	Type	Size	Key
Volunteer_ID	Unique volunteer ID	Int	4 Bytes	Primary
Name	Volunteer name	String	255 Bytes	No
Availability	Time available	String	255 Bytes	No
Profile_Details	Bio or skills	Text	Unlimited	No

Volunteer Hours

Attribute	Definition	Type	Size	Key
Entry_ID	Unique entry ID	Int	4 Bytes	Primary
Volunteer_ID	Volunteer reference	Int	4 Bytes	Foreign
Hours	Logged hours	Float	4 Bytes	No
Date	Date of service	Date	8 Bytes	No

Volunteer Event

Attribute	Definition	Type	Size	Key
Volunteer_ID	Volunteer reference	Int	4 Bytes	Foreign
Event_ID	Event reference	Int	4 Bytes	Foreign
Registered_Date	Date of registration	Date	8 Bytes	No

Event

Attribute	Definition	Type	Size	Key
Event_ID	Unique event ID	Int	4 Bytes	Primary
Name	Event name	String	255 Bytes	No
Date	Event date	Date	8 Bytes	No
Time	Event time	Time	8 Bytes	No
Location	Event location	String	255 Bytes	No
Registration_Fee	Fee for participation	Float	4 Bytes	No

Customer Support Staff

Attribute	Definition	Type	Size	Key
Support_ID	Unique support staff ID	Int	4 Bytes	Primary

Name	Name of the staff member	String	255 Bytes	No
Email	Email address	String	255 Bytes	No

Ticketing System

Attribute	Definition	Type	Size	Key
Ticket_ID	Unique ticket ID	Int	4 Bytes	Primary
User_ID	Reference to User	Int	4 Bytes	Foreign
Issue	Description of the issue	Text	Unlimited	No
Status	Current ticket status	String	50 Bytes	No

ChatBot

Attribute	Definition	Type	Size	Key
Bot_ID	Unique bot ID	Int	4 Bytes	Primary
Bot_Name	Chatbot name	String	100 Bytes	No
Purpose	Function of the chatbot	String	255 Bytes	No

FAQ

Attribute	Definition	Type	Size	Key
FAQ_ID	Unique FAQ ID	Int	4 Bytes	Primary
Question	Frequently asked question	Text	Unlimited	No
Answer	Answer to the question	Text	Unlimited	No

System Admin

Attribute	Definition	Type	Size	Key
SysAdmin_ID	Unique system admin ID	Int	4 Bytes	Primary
Name	Admin's name	String	255 Bytes	No
Email	Email address	String	255 Bytes	No

Administrator

Attribute	Definition	Type	Size	Key
Admin_ID	Unique admin ID	Int	4 Bytes	Primary
Name	Admin's name	String	255 Bytes	No
Email	Email address	String	255 Bytes	No
Access_Lvl	Permission level	Int	4 Bytes	No

Content

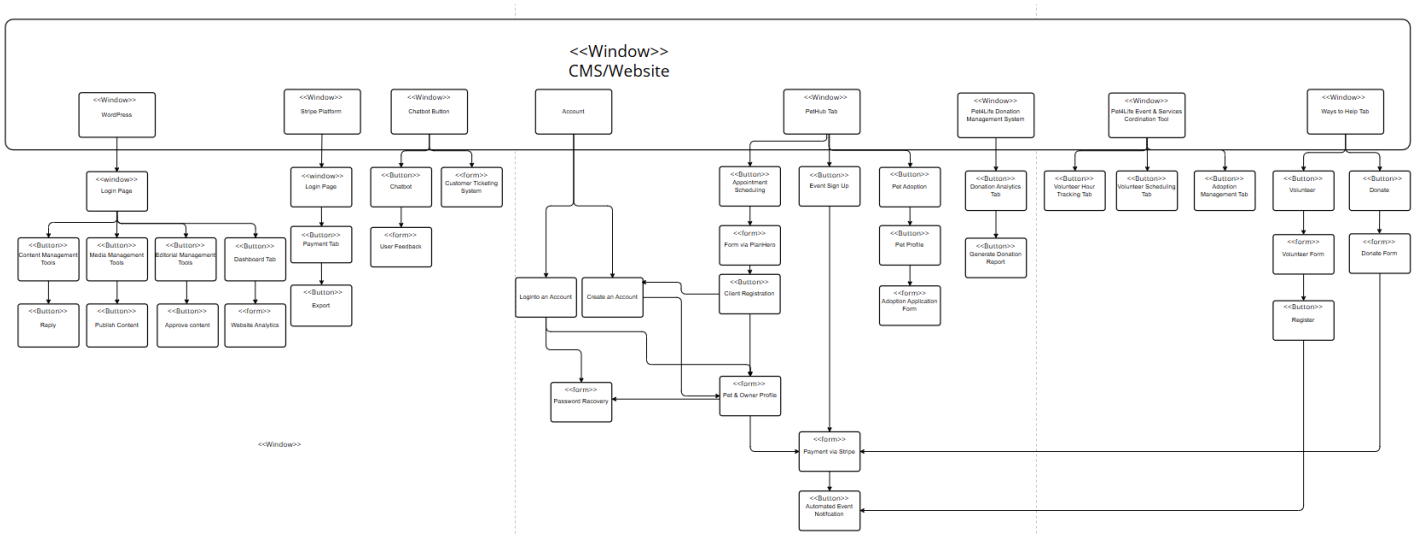
Attribute	Definition	Type	Size	Key
Content_ID	Unique content ID	Int	4 Bytes	Primary
Title	Content title	String	255 Bytes	No
Body	Main content body	Text	Unlimited	No
Created_By	Creator's employee ID	Int	4 Bytes	Foreign

ContentMGM Systems

Attribute	Definition	Type	Size	Key
CMS_ID	Unique system ID	Int	4 Bytes	Primary
System_Name	Name of the CMS	String	100 Bytes	No
Version	Version of the CMS	String	50 Bytes	No

3. User Interface Design

3.1 User Interface Navigation Diagram



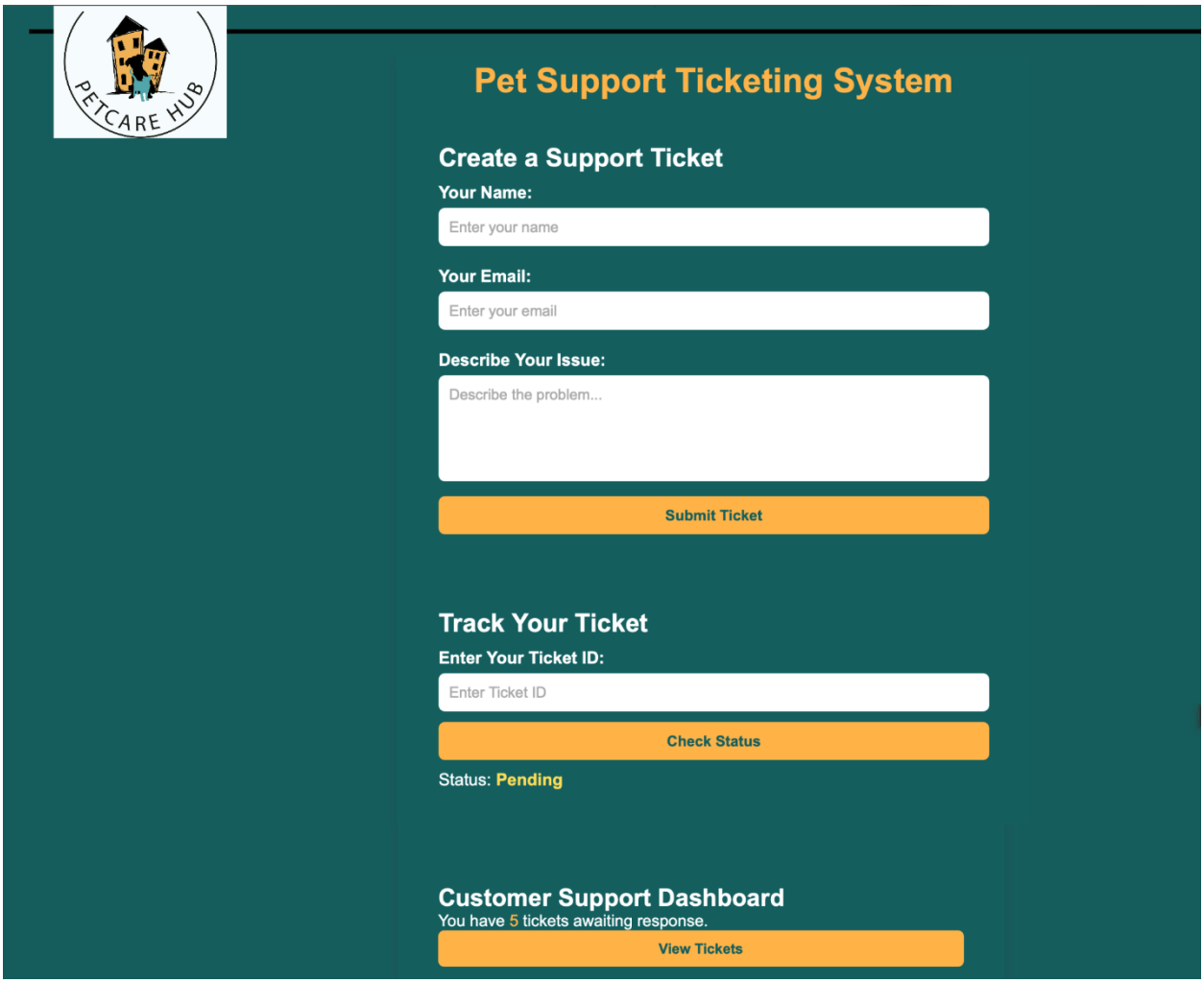
3.2 User Interface Diagram Narrative

This is a User Interface Diagram that depicts the connection between different parts of the proposed Pet4Life system. These connections will explain how different users will interact with the system and how separate parts of the system will interact with each other.

3.3 Screen Layouts

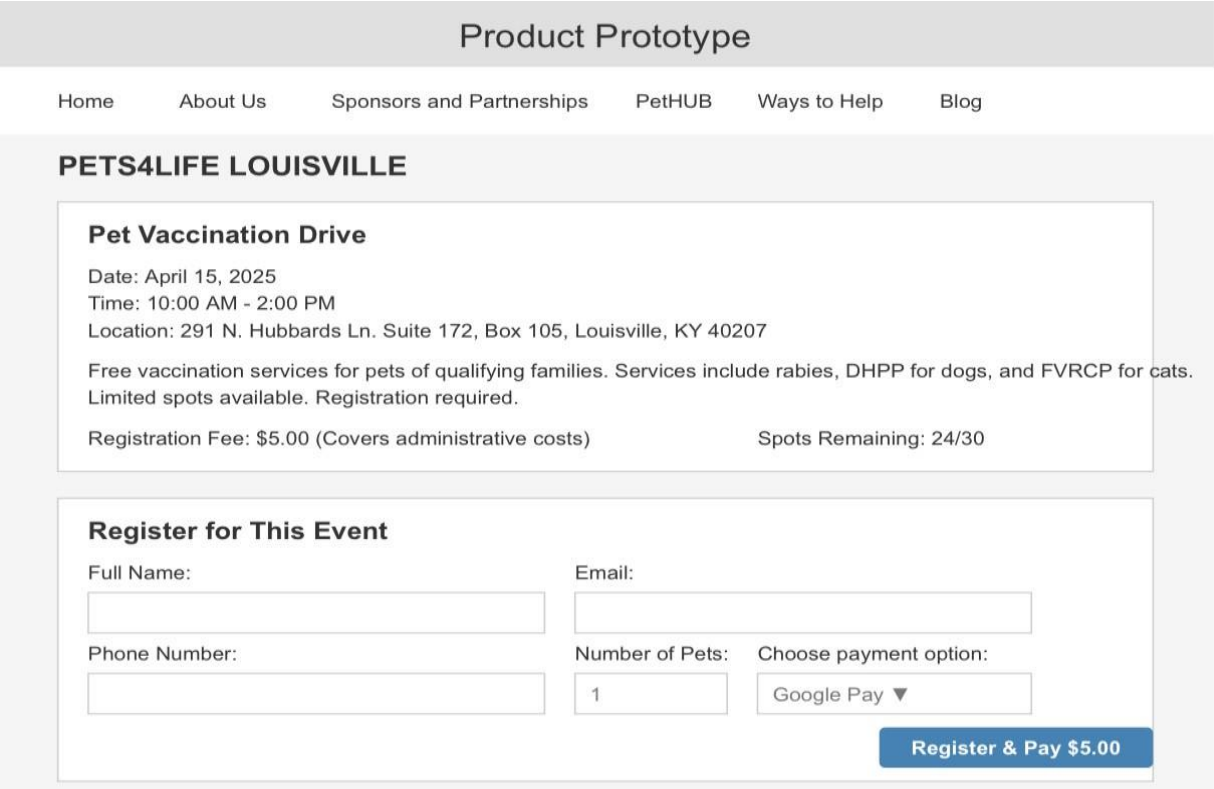
3.3.A Data Capture Forms

Fill out a support ticket



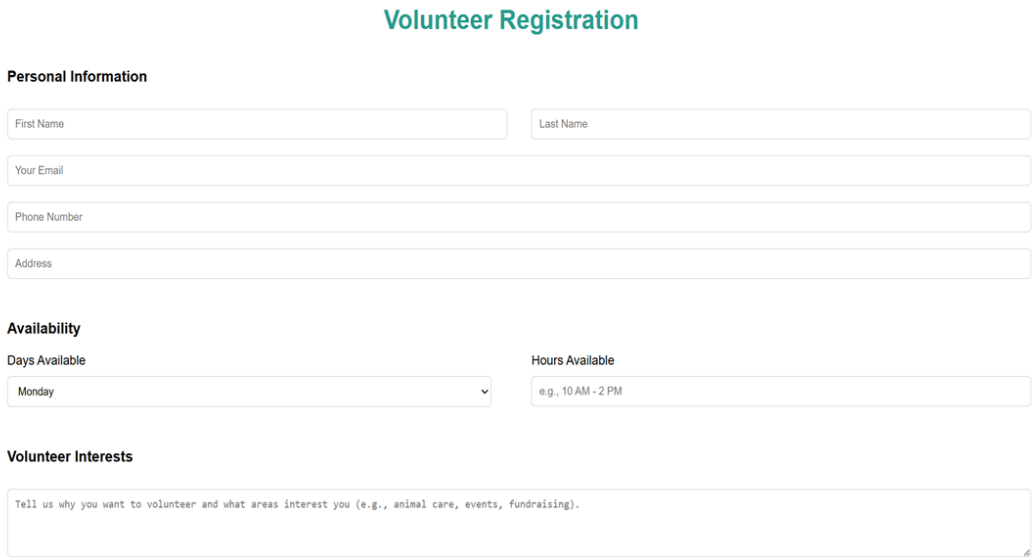
This screenshot shows how the pet support ticketing form will work. It will include data entry for the owner's name, email address, text box to describe their problem and how Pet4Life can help them. It will also include an entry for ticket number to view the status on it and a button to see all the tickets that a user has submitted.

Event Registration and Payment



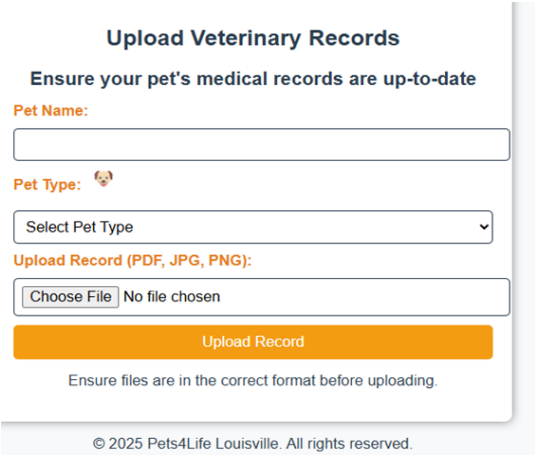
The user will click on the PetHub tab on Pet4Life's website which will contain information about upcoming events that Pet4Life is hosting. The user will then have the option to register for the event and pay the upfront fee for the service that Pet4Life will offer at the event.

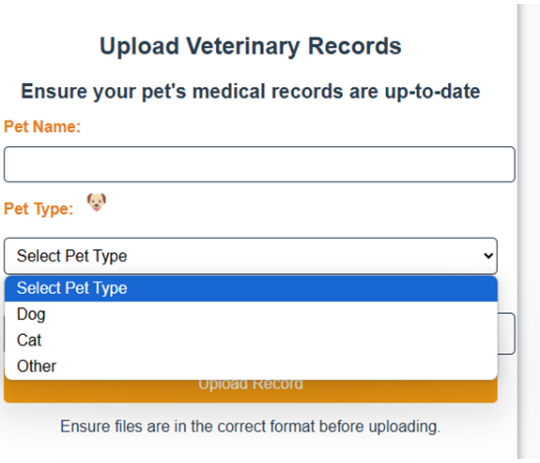
Volunteer Registration



The user will click on the "Ways to help" tab on Pet4Life's website and click on the "Volunteer" button. This will open a form for the user to fill out with their personal information, contact information, availability, and interests. The user will click on the register button to submit their form and may be redirected to automated event notification page.

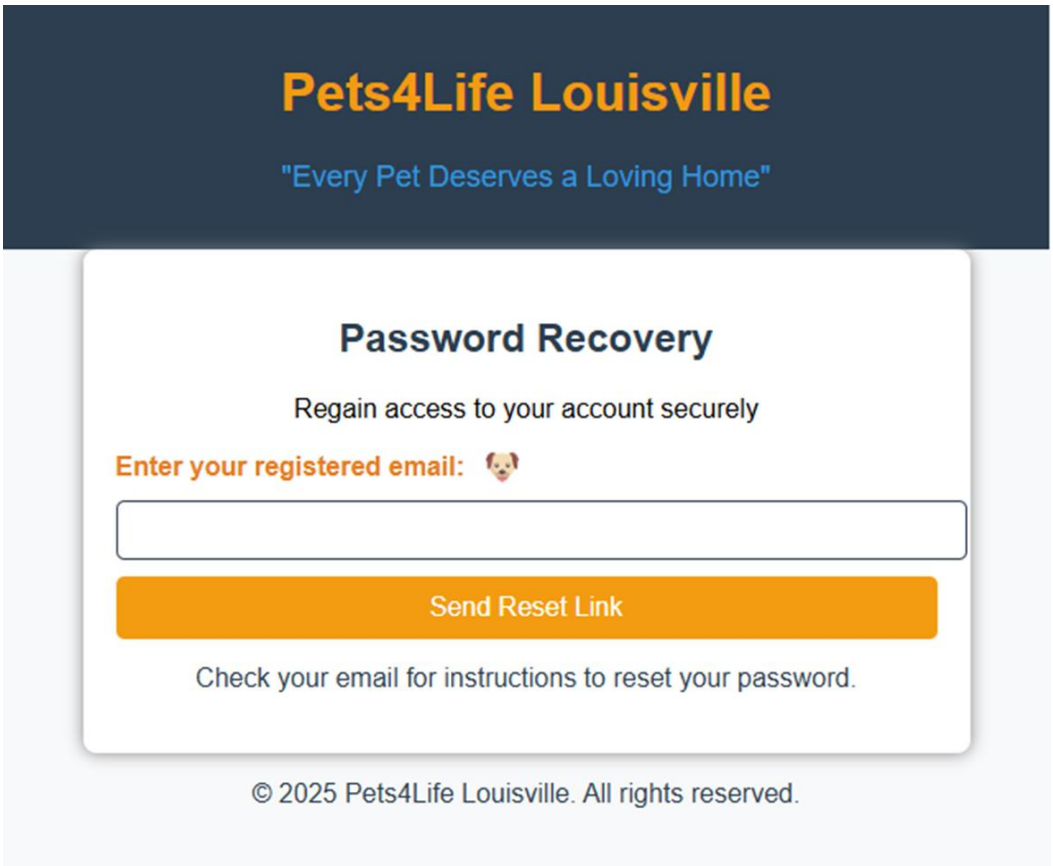
Upload Veterinary Records





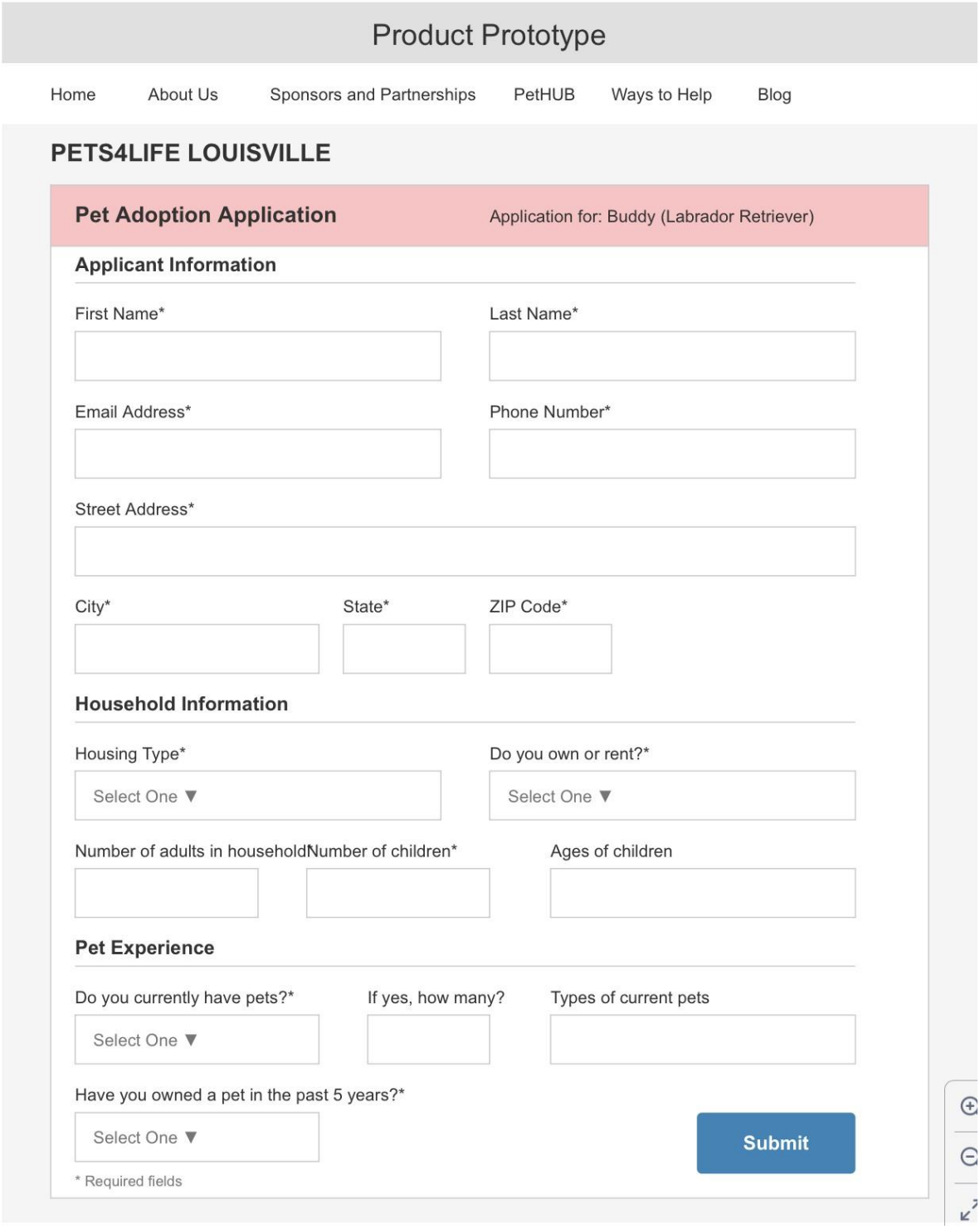
This screenshot depicts the form for uploading veterinary records for the pet by the owner. The pet owner will fill out this form and attach required vet records when scheduling an appointment. They will click on the “Schedule appointment” button under the “PetHub” tab on the Pet4Life website.

Password Recovery



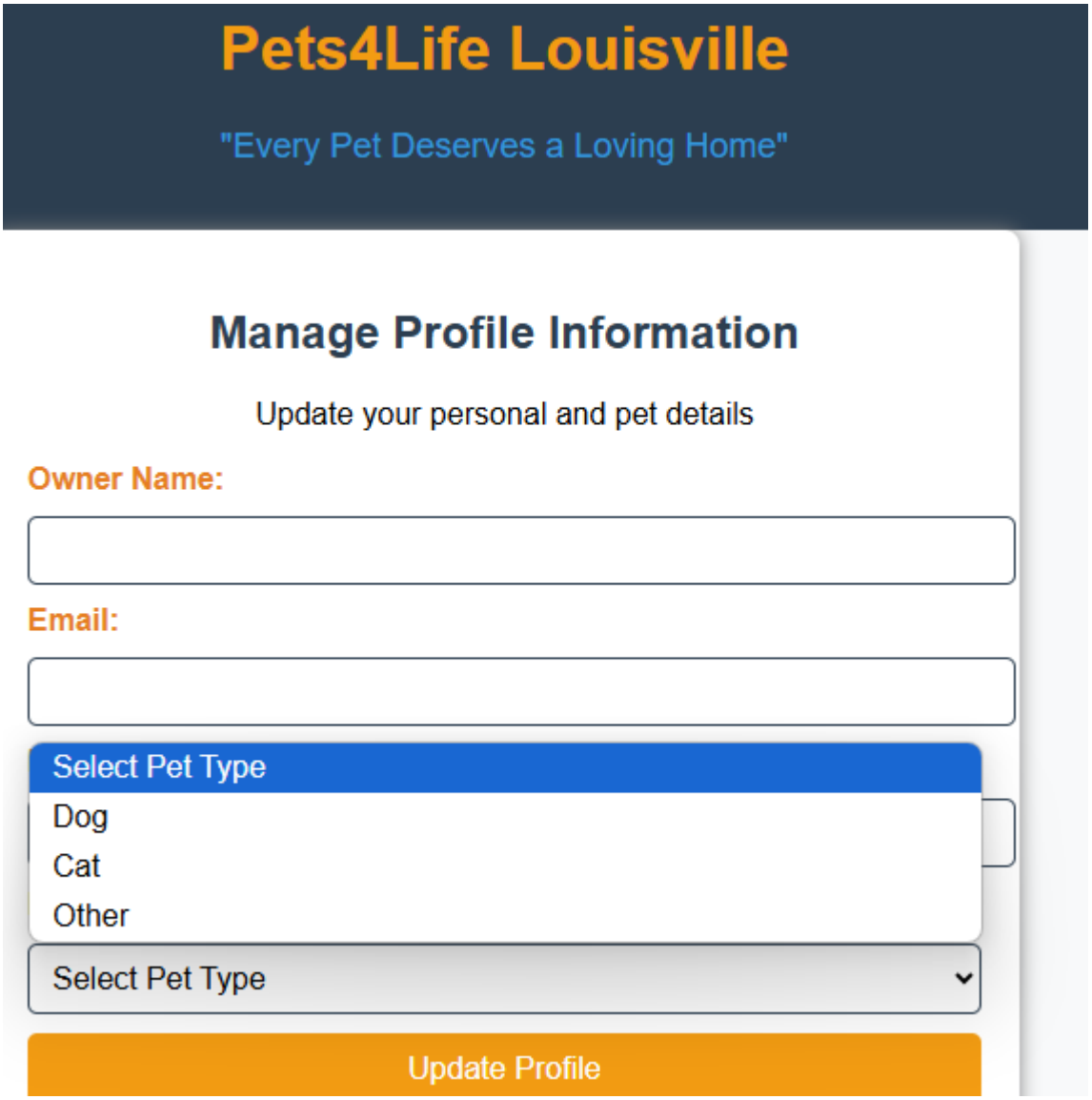
This screenshot depicts the small form for when a user has forgotten their password to their account and chooses to recover their lost password. The user will be given the option to recover their password when filling out forms. The user will click on the

Pet Adoption Form



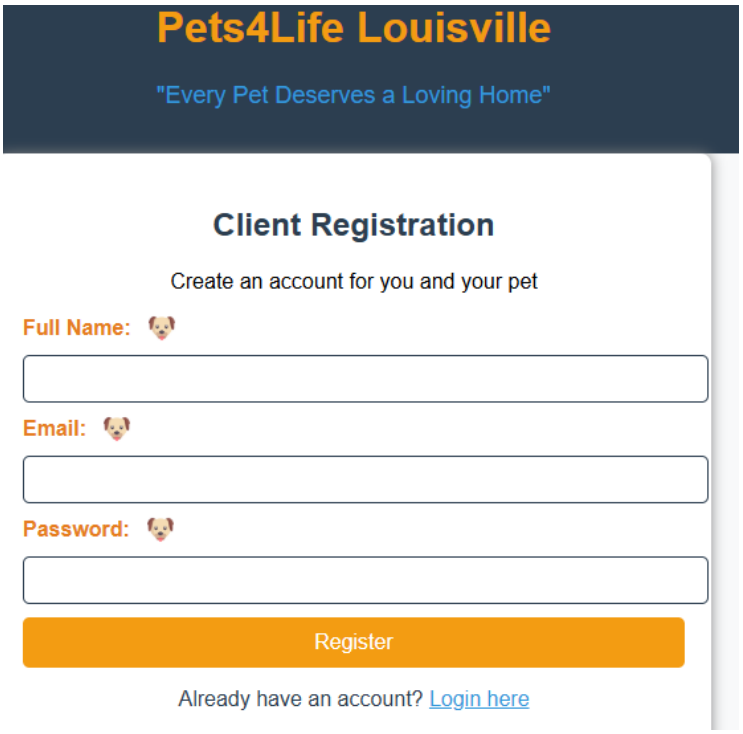
This screenshot depicts the form that any user wishing to adopt a pet will have to fill out in order to be considered for the adoption of a pet. They will click on the PetHub tab on the Pet4Life website and then click on the adoption tab. They will then be greeted with pet profiles for pets that are in the adoption system. They will choose a pet and fill out a form.

Profile Management



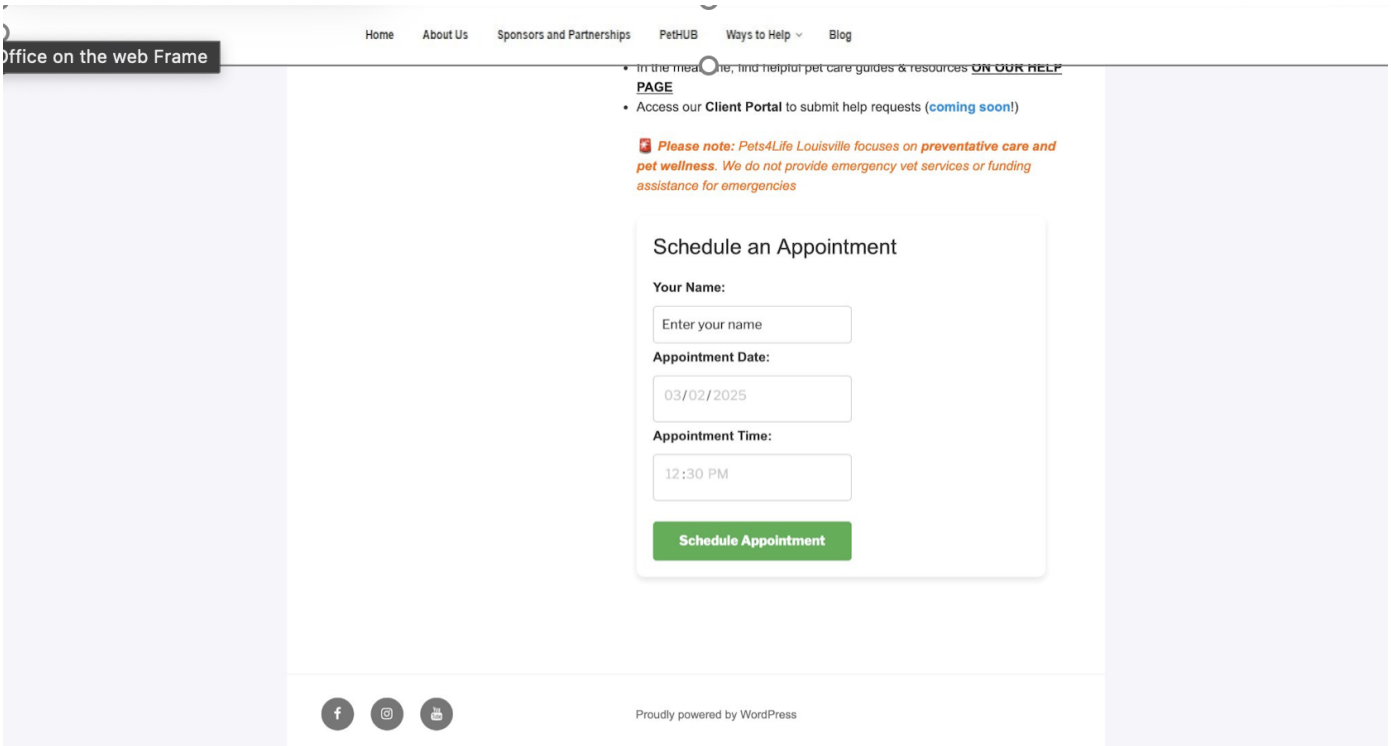
This screenshot depicts the profile management page for the user after creating an account. This page will be available after registering an account after scheduling an appointment or logging into from the homepage or scheduling an appointment with an existing account. The user will click on the Pet & Owner profile button after clicking logging in and entering their credentials or after creating an account.

Client Registration and Login



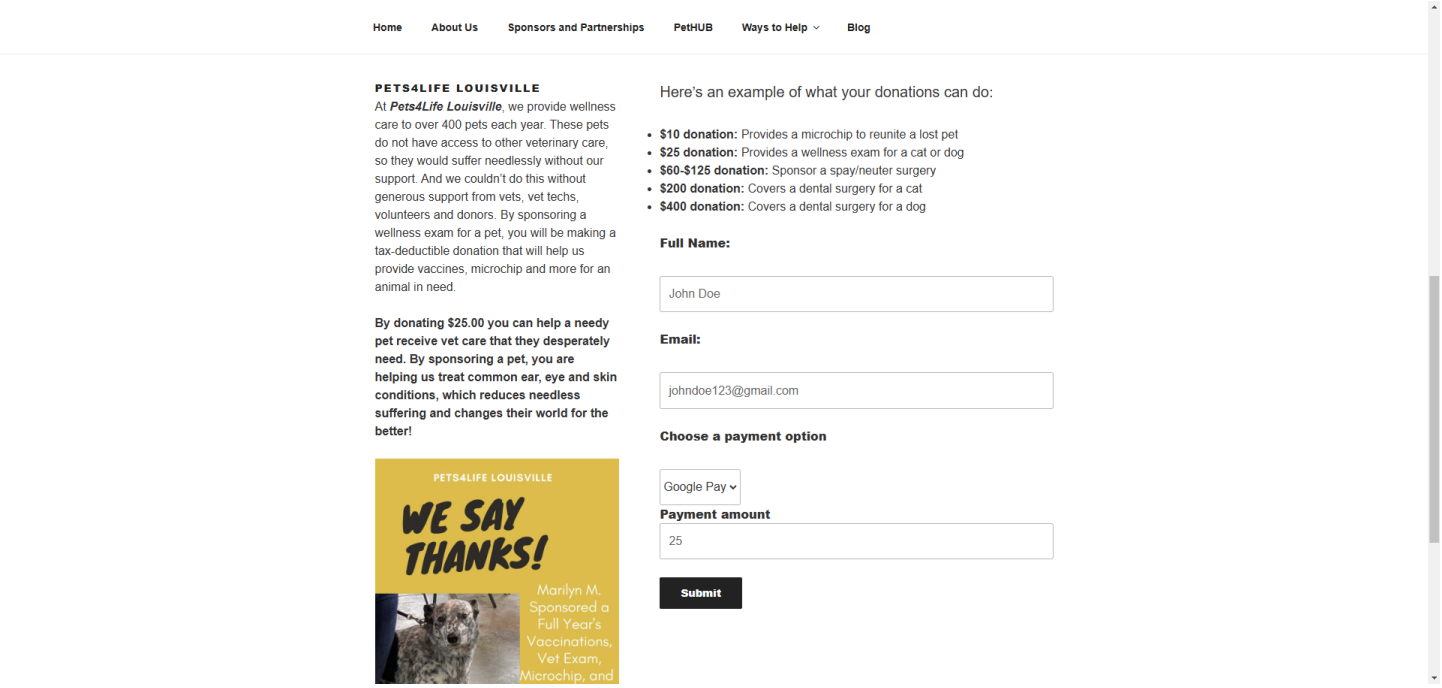
This screenshot depicts the form that the user will fill out when making an account during appointment scheduling on Pet4Life's website or logging into their already existing acount. They will click the PetHub tab, click the schedule appointment tab. Upon finishing the schedule appointment form, they will be greeted to register an account to hold their personal info and pet info. They can also access it from the Account tab and then clicking on the login button.

Scheduling an Appointment



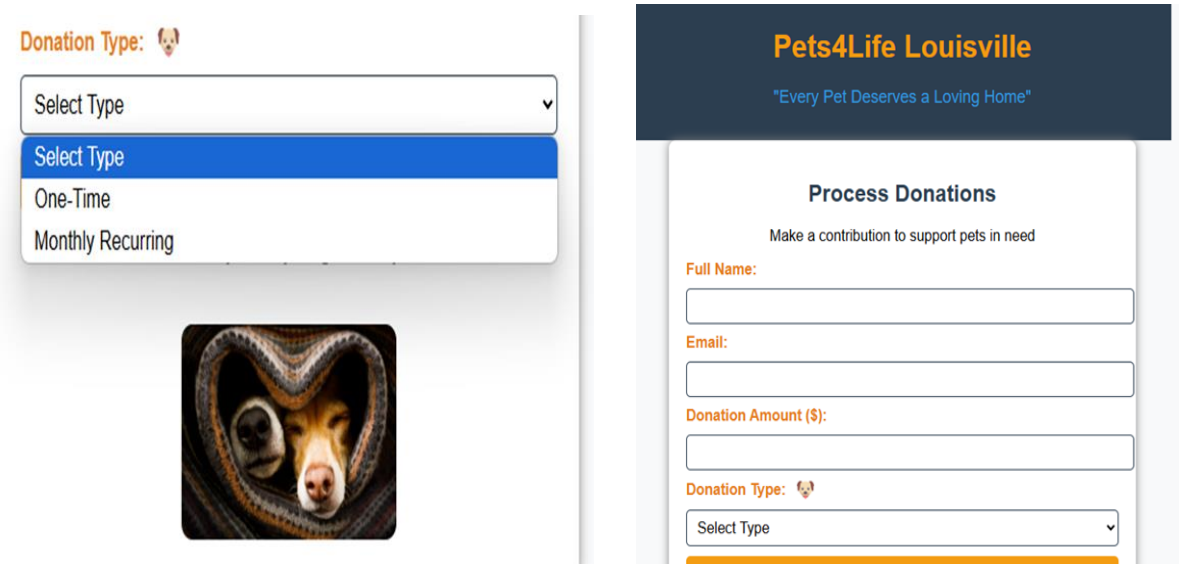
This screenshot depicts the form that users will fill out when creating an appointment. Users will click on the PetHub tab on the Pet4Life website, click on the schedule appointment tab which will direct them to this form to fill out.

Payment via Stripe Form



This screenshot depicts the form that will be completed when doing any sort of payment on the Pet4Life website. In this example, the user will be greeted with a form hosted by Stripe that will collect payment info and initiate the transaction after completing the donation form.

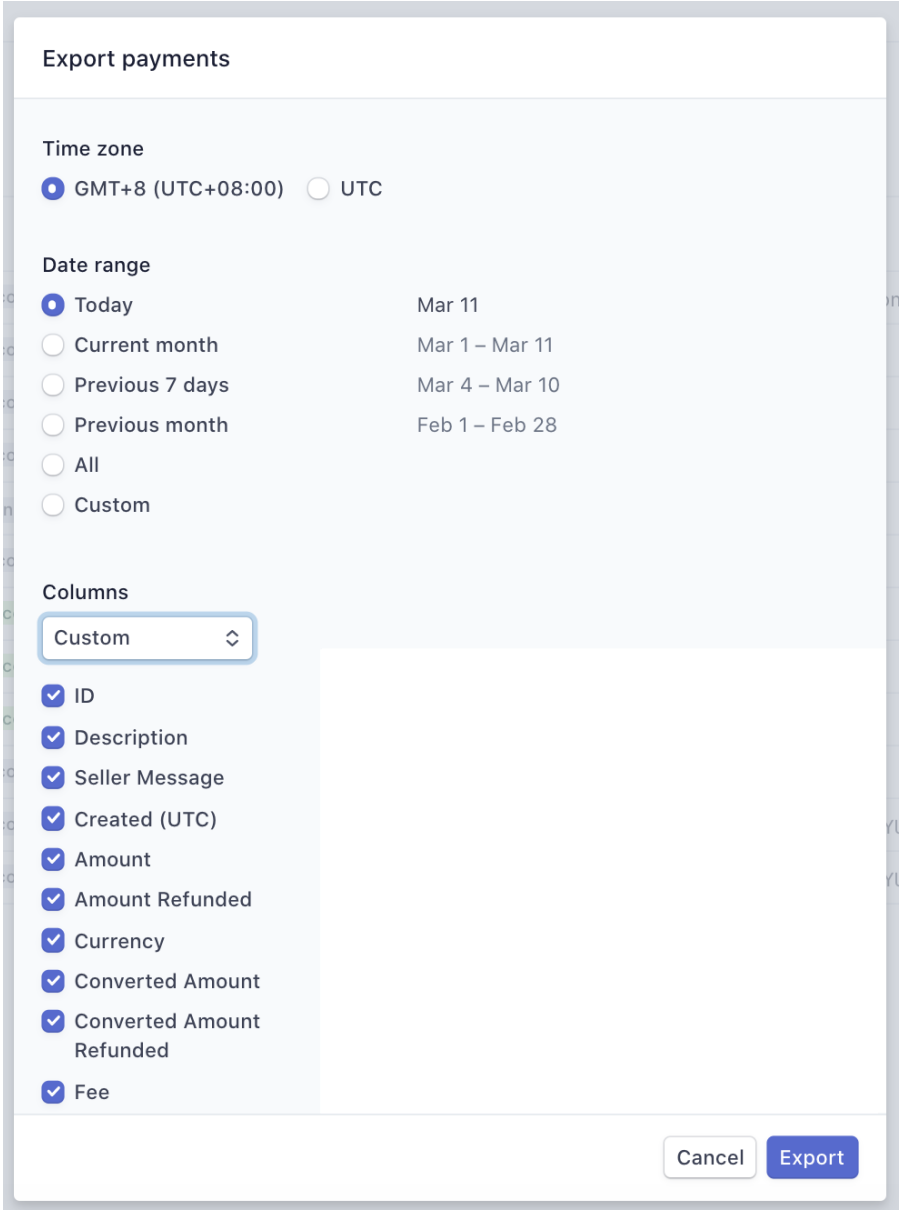
Donation Form



This screenshot depicts the form that the user will fill out when donating. The user will visit the Pet4Life website, click on the Ways to help tab, and then click on the Donate button to be directed to this form.

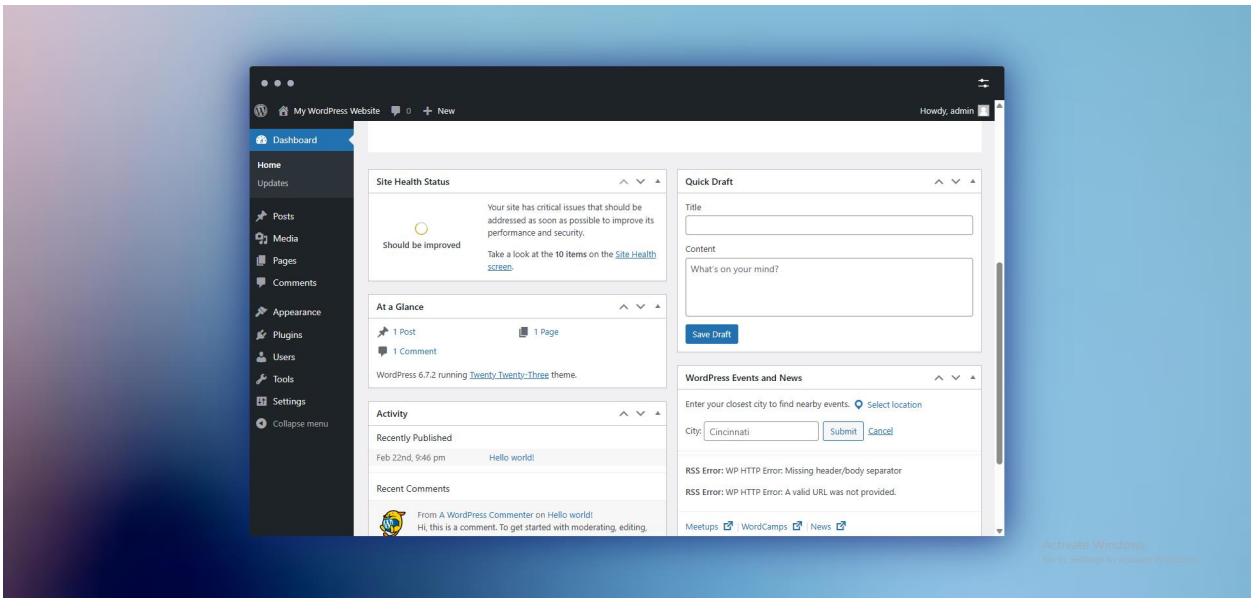
3.3.B Data Presentation Forms

Access financial report



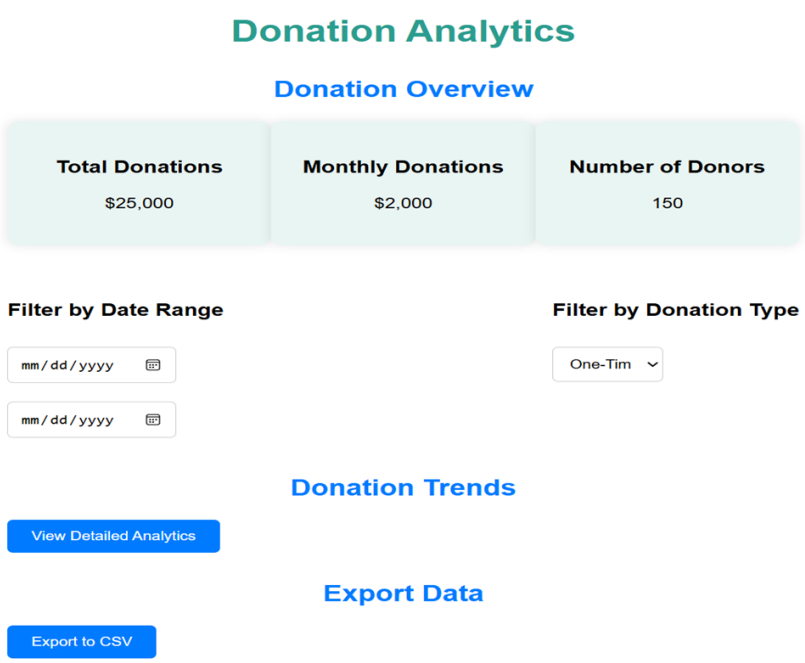
The system administrator will log into the company Stripe account, click on the payment tab, and click on the export button which will open a “Export payments” window where the system administrator will be able to download a file with transaction data. This transaction data can also be sorted through provided criteria to download specific and relevant data.

Access Website Analytics



This screenshot depicts the information provided after accessing the analytics of activities and statistics of the Pet4Life website. The System Administrator will login to the WordPress CMS, they will click on the dashboard tab and will be able to see website analytics.

Donation Analytics

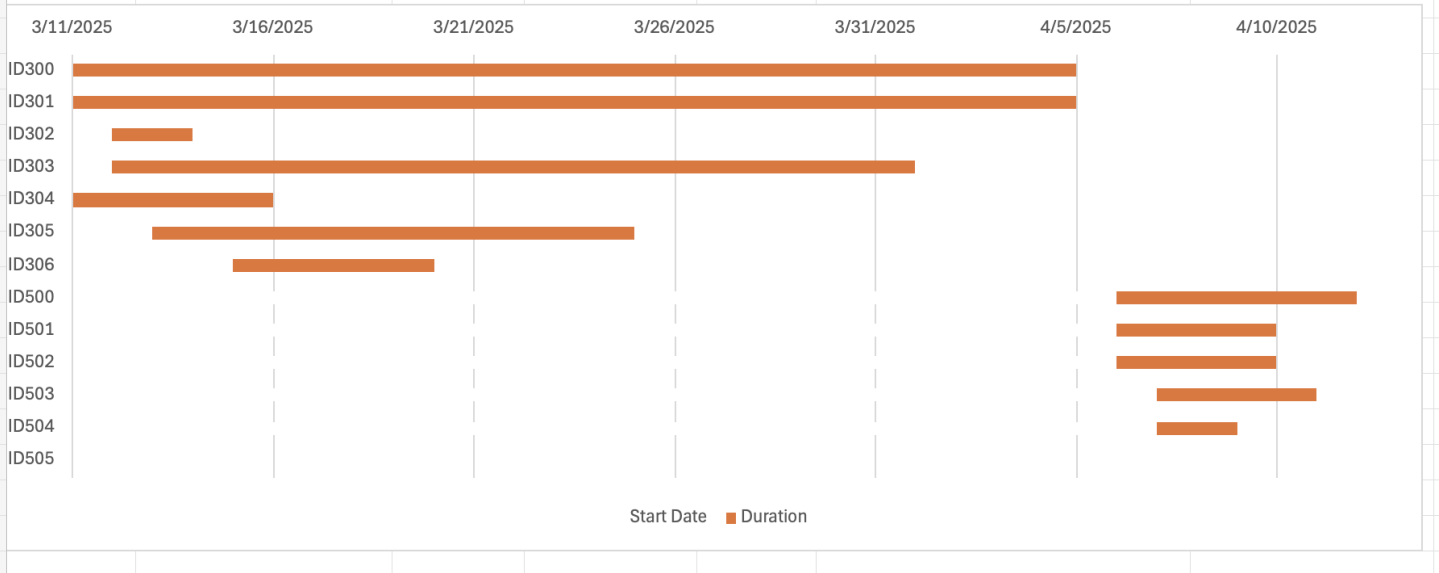


This screenshot depicts the information that will be provided when viewing the donation analytics page. The Fundraising manager will be on the admin page of the Pet4Life website and from transactional data from Stripe. The Fundraising manager will click on the donation management tab, click on the generated report button, and they will view the information as shown.

4. Gantt Chart

4.1 Gantt Chart Diagram

Task ID	Task	Start Date	End date	Duration	Task Responsibilty	Dependency
ID300	Gantt Chart	3/11/2025	4/5/2025	25	Kyah Owens	
ID301	I3 Document Compile	3/11/2025	4/5/2025	25	Isaiah Martinez	
ID302	System Requirements	3/12/2025	3/14/2025	2	Eric Cato	
ID303	Use case	3/12/2025	4/1/2025	20	Everyone	ID302
ID304	Trace Matrix	3/11/2025	3/16/2025	5	Kyah Owens	ID302, ID303
ID305	Use Case Diagram	3/13/2025	3/25/2025	12	Everyone	ID302
ID306	Use Case Prototype	3/15/2025	3/20/2025	5	Everyone	
ID500	Gantt Chart	4/6/2025	4/12/2025	6	Kyah Owens	
ID501	Database Design	4/6/2025	4/10/2025	4	Eric Cato and Greiler Abrahantes	ID504
ID502	Database Definition	4/6/2025	4/10/2025	4	Eric Cato and Greiler Abrahantes	ID504
ID503	Screen Layout	4/7/2025	4/11/2025	4	Everyone	
ID504	Class Diagram	4/7/2025	4/9/2025	2	Ozine	
ID505	User Interface Nav	4/6/2020	4/12/2025	6	Isaiah Martinez	



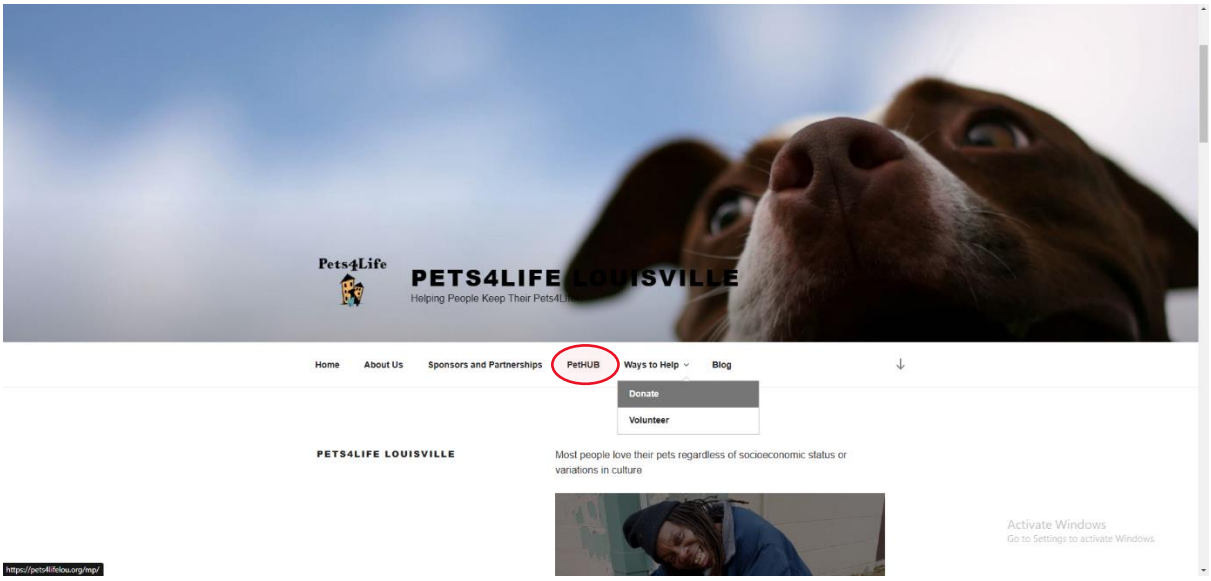
4.2 Gantt Chart Narrative

The project starts with the Gantt chart and Document creation in 25 days, starting 3/11/2025 (ID300, ID301). System Requirements (ID302) will be completed by 3/15/2025; these will specify the use case (ID303) system which will be done by 4/1/2025. Matrix (ID304) will take 5 days and 12 days for Use Case Diagram (ID305), both depending on System Requirements (ID302) and Use Case (ID303).

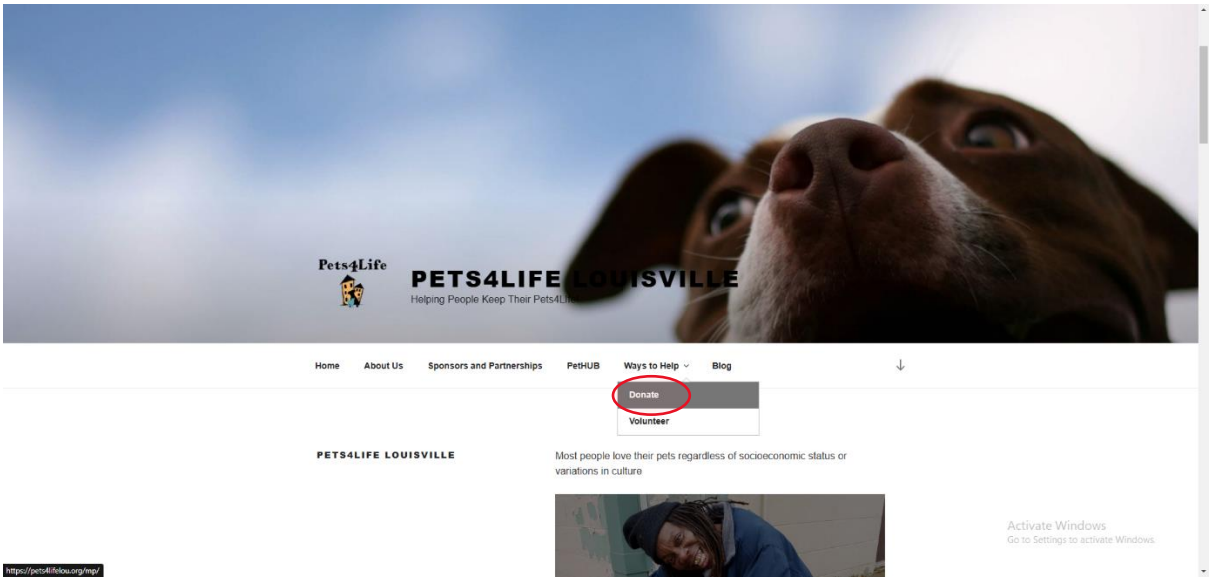
Gantt Chart (ID500) update will be in the next phase. Database Design (ID501) and Definition (ID502) take 4 days to complete each . Additionally, Screen Layout (ID503) will need 4 days as well. All these depend on Class Diagram (ID504), which takes 2 days . In the final step, User Interface Navigation will yield 6 days.

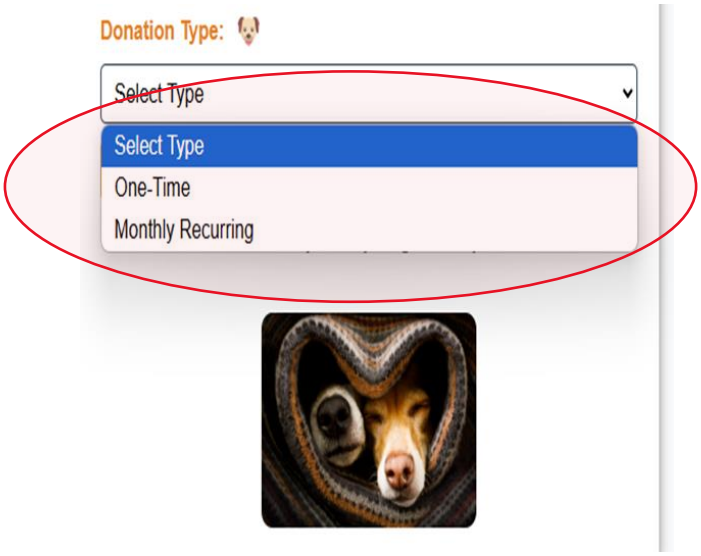
5. User Interface Prototypes

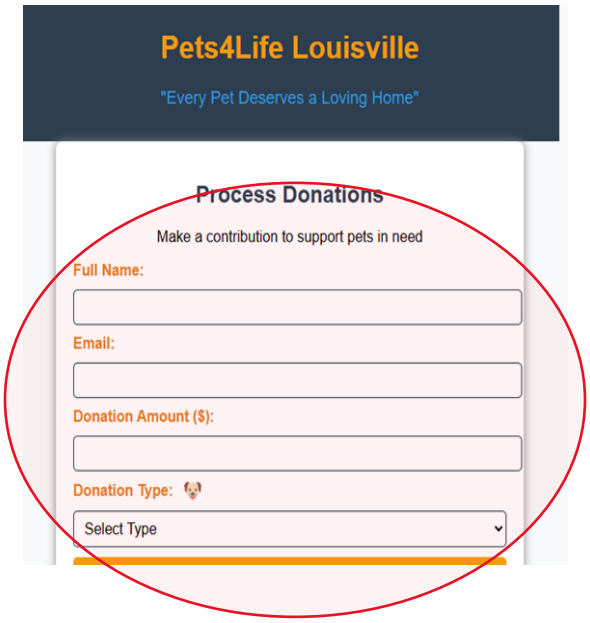
Use Cases 1 - 4



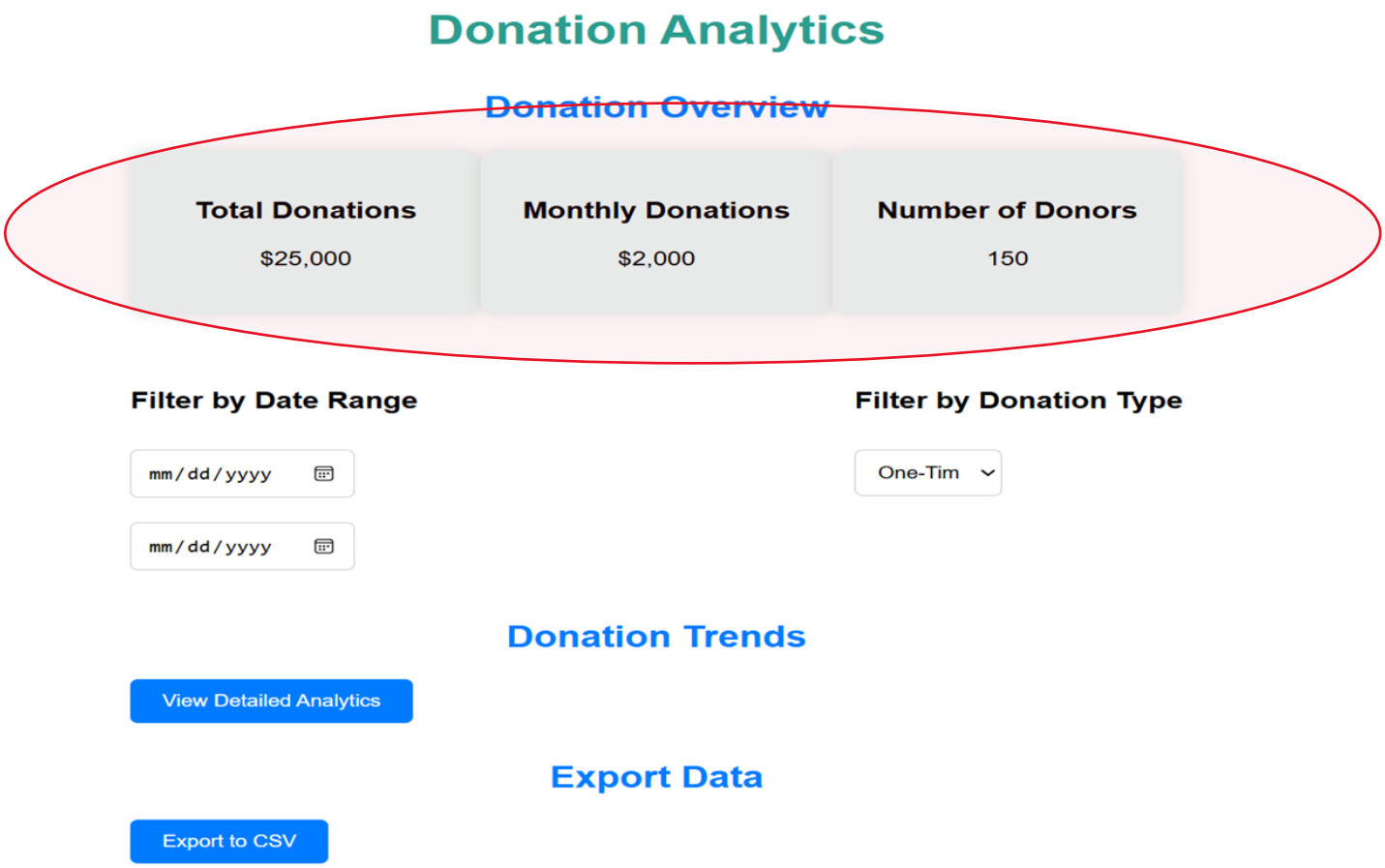
Use Cases 5 - 7


Pets4Life
PETS4LIFE LOUISVILLE
Helping People Keep Their Pets4Life
Home
About Us
Sponsors and Partnerships
PetHUB
Ways to Help
Blog
Donate
Volunteer
PETS4LIFE LOUISVILLE
Most people love their pets regardless of socioeconomic status or variations in culture
Activate Windows
Go to Settings to activate Windows.
https://pets4lifelou.org/mp/

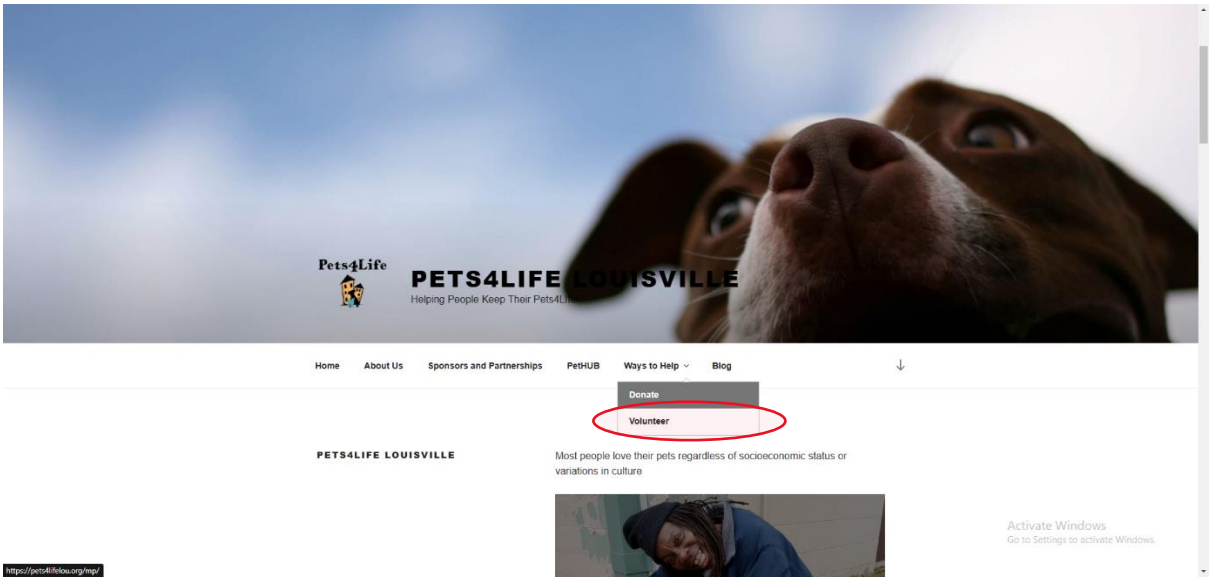

Donation Type:
Select Type
Select Type
One-Time
Monthly Recurring


Pets4Life Louisville
"Every Pet Deserves a Loving Home"
Process Donations
Make a contribution to support pets in need
Full Name:
Email:
Donation Amount ($):
Donation Type:
Select Type

Use Cases 8



Use Cases 9 – 11



Use Case 12

Volunteer Tracking

Volunteer: John Doe

Email: john.doe@email.com

Phone: (555) 123-4567

Volunteer Role: Event Coordinator

Volunteer Hours Logged

Date	Event	Hours Worked	Notes
March 5, 2025	Pet Adoption Event	5	Assisted with event setup
March 10, 2025	Volunteer Orientation	2	Helped with training new volunteers
March 12, 2025	Fundraising Event	6	Coordinated donation collection

Log New Volunteer Hours

Event Name

mm/dd/yyyy

Volunteer Tracking

Volunteer: John Doe

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Volunteer Role: Event Coordinator

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Log New Volunteer Hours

Event Name

mm/dd/yyyy

Use Case 13

Product Prototype

Home | About Us | Sponsors and Partnerships | PetHUB | Ways to Help | Blog

PETS4LIFE LOUISVILLE - STAFF DASHBOARD

Staff-only page: Create and manage events for Pets4Life Louisville

Event Title:
Pet Vaccination Drive

Event Date: 04/15/2025
Start Time: 10:00 AM
End Time: 2:00 PM

Location:
291 N. Hubbards Ln. Suite 172, Box 105, Louisville, KY 40207

Event Description:
Free vaccination services for pets of qualifying families.
Services include rabies, DHPP for dogs, and FVRCP for cats.
Limited spots available. Registration required.

Registration Fee (optional): $5.00
Maximum Participants: 30

Create Event

Use Case 14 – 16

Product Prototype

Home | About Us | Sponsors and Partnerships | PetHUB | Ways to Help | Blog

PETS4LIFE LOUISVILLE

Pet Vaccination Drive

Date: April 15, 2025
Time: 10:00 AM - 2:00 PM
Location: 291 N. Hubbards Ln. Suite 172, Box 105, Louisville, KY 40207

Free vaccination services for pets of qualifying families. Services include rabies, DHPP for dogs, and FVRCP for cats.
Limited spots available. Registration required.

Registration Fee: $5.00 (Covers administrative costs)
Spots Remaining: 24/30

Register for This Event

Full Name:
Email:
Phone Number:
Number of Pets: 1
Choose payment option: Google Pay ▼

Register & Pay $5.00

Product Prototype

Home | About Us | Sponsors and Partnerships | PetHUB | Ways to Help | Blog

PETS4LIFE LOUISVILLE - NOTIFICATION SYSTEM

Event: Pet Vaccination Drive

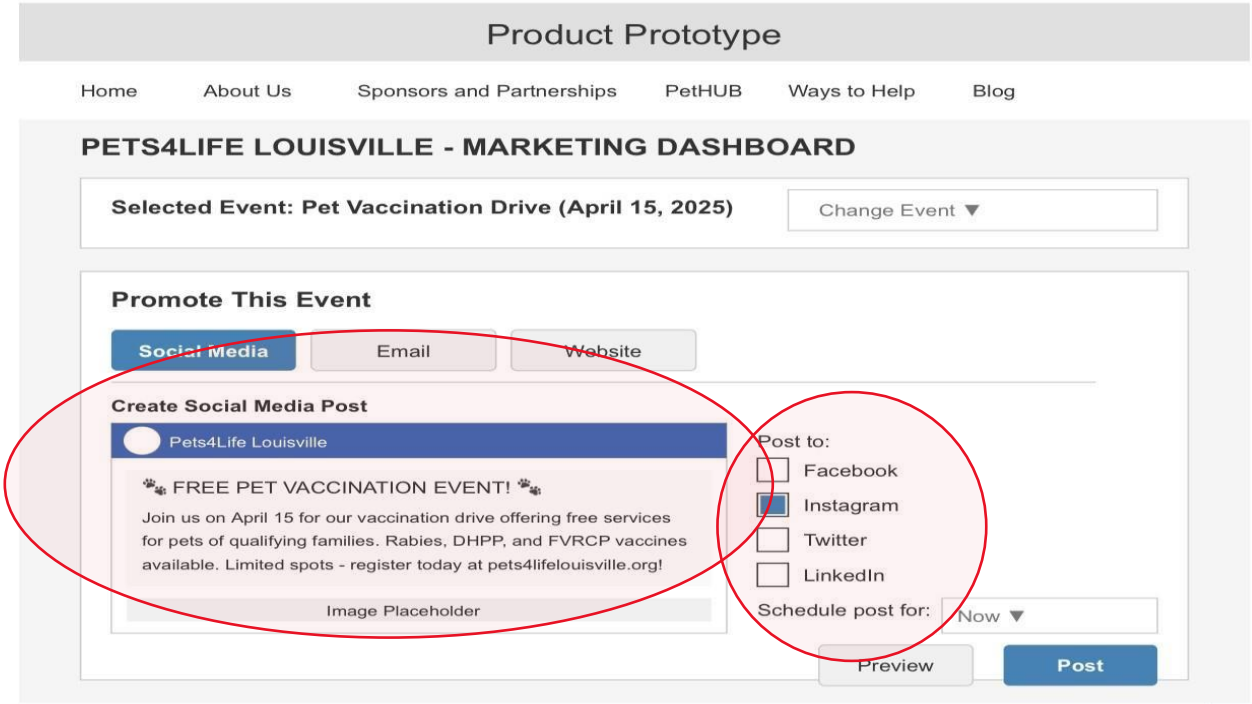
Date: April 15, 2025 | Time: 10:00 AM - 2:00 PM
Total Registrants: 6 participants

Automatic Notifications:

Notification Type	Timing	Status	Actions
Registration Confirmation	Immediate after registratic	Active	Edit Template
1-Week Reminder	April 8, 2025 (10:00 AM)	Active	Edit Template
1-Day Reminder	April 14, 2025 (10:00 AM)	Active	Edit Template
Post-Event Survey	April 15, 2025 (4:00 PM)	Active	Edit Template
Custom Notification	Manual Trigger	Inactive	Create & Send

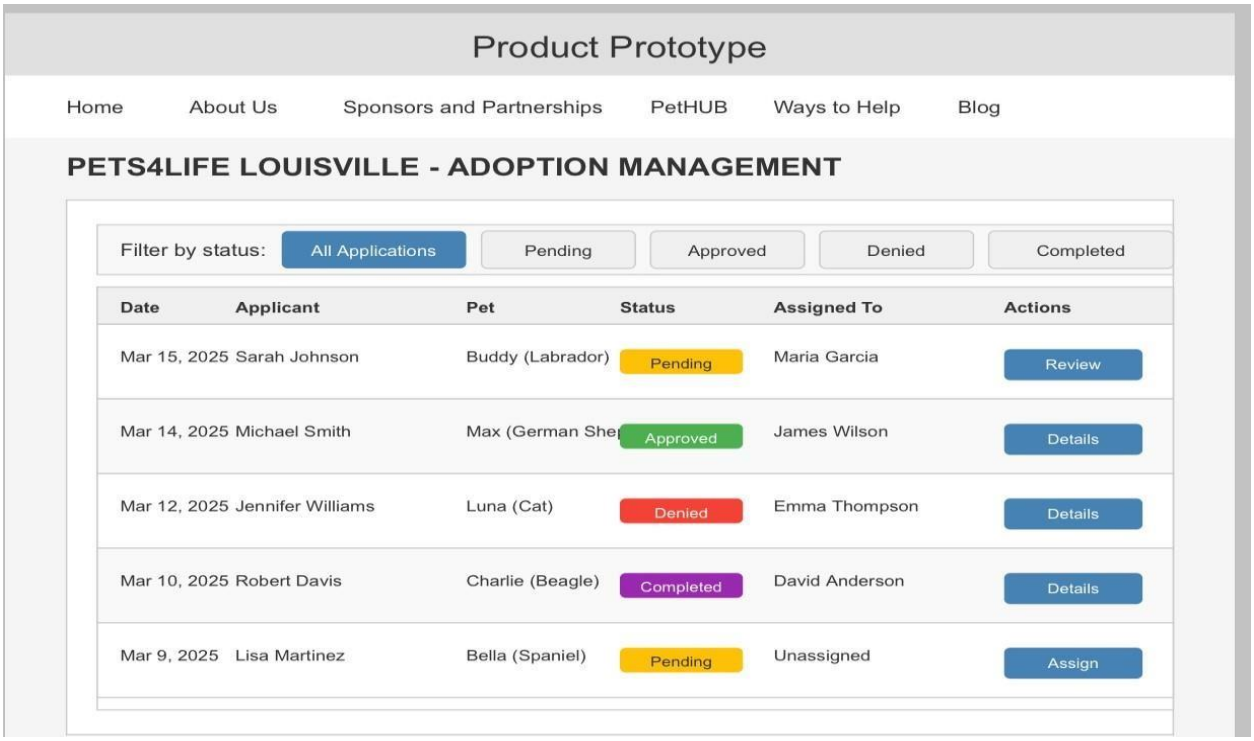
Test Notification System: Enter test email address | Send Test

Use Case 17

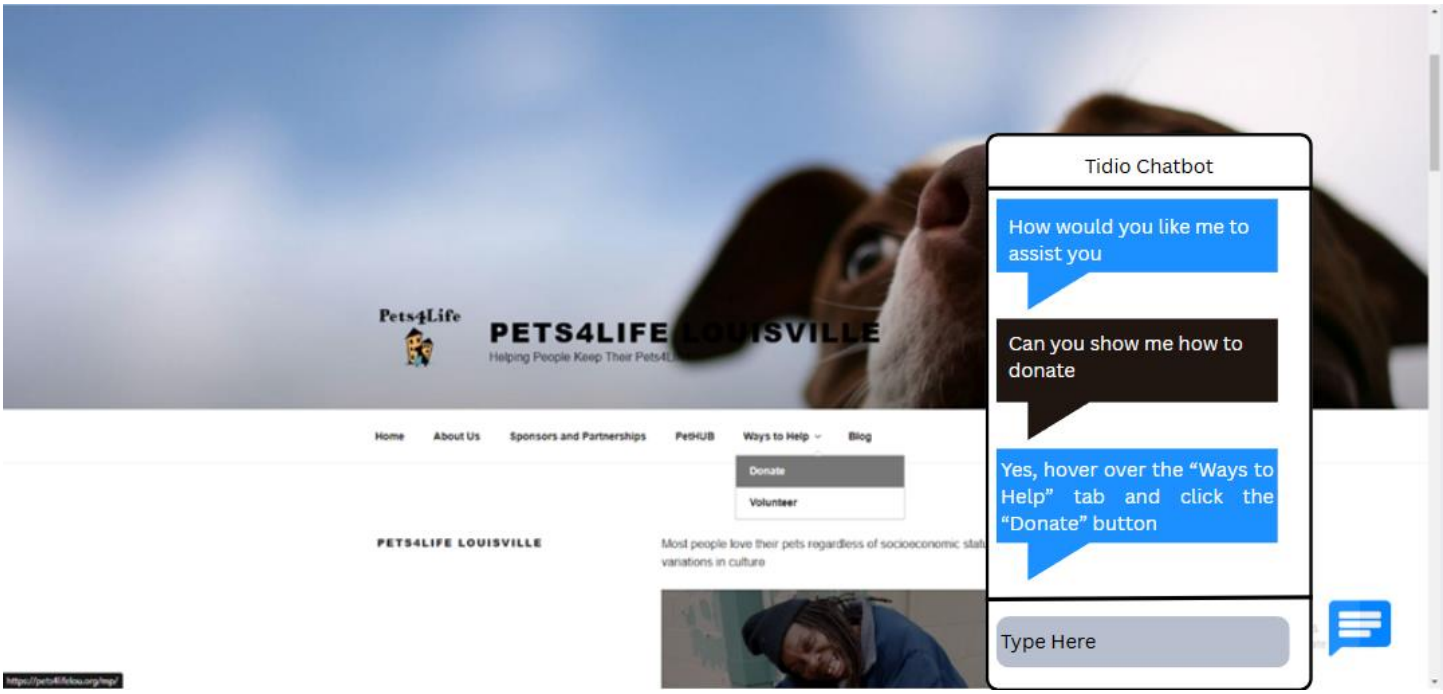


Use Case 18

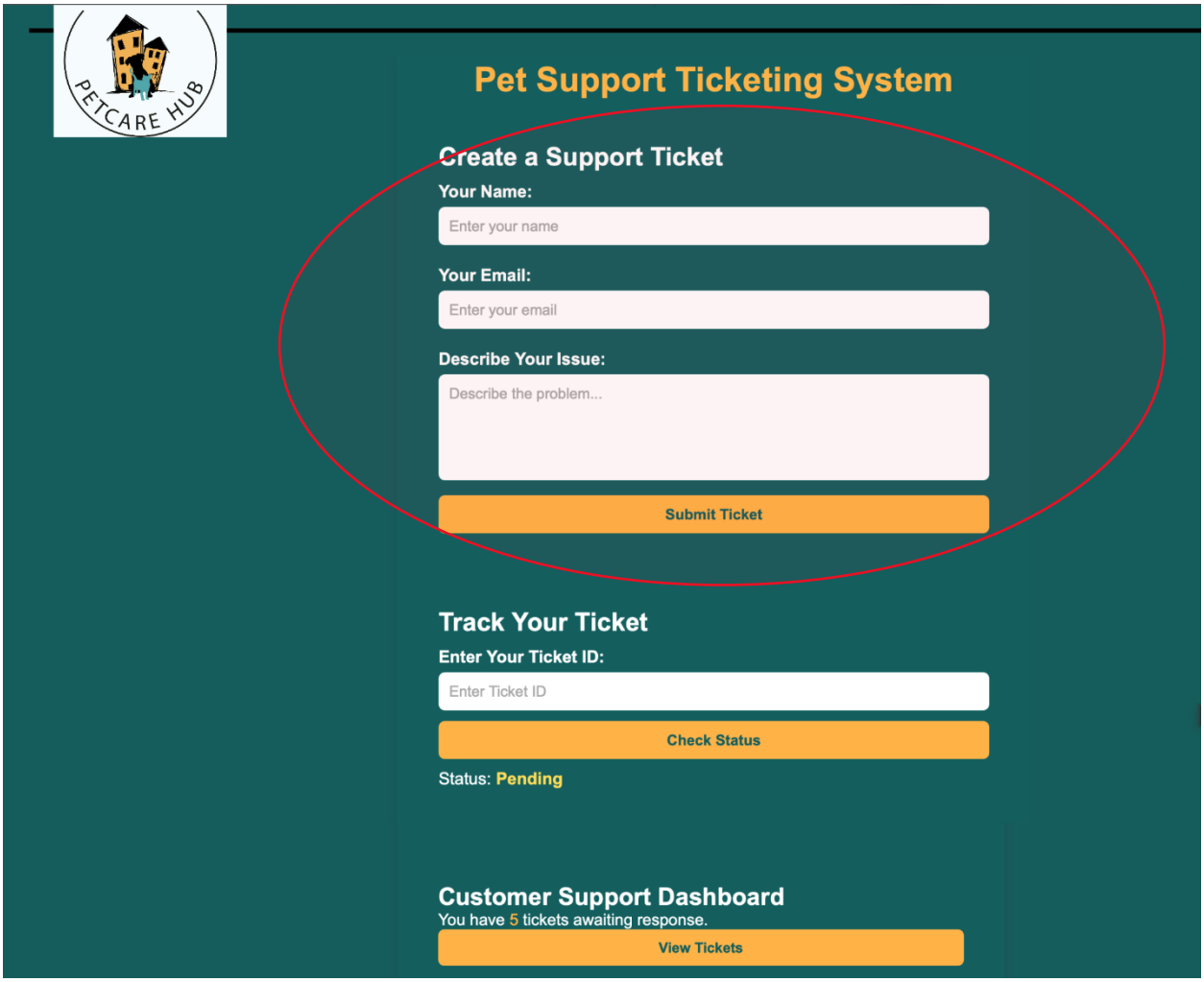
Use Case 19 - 20



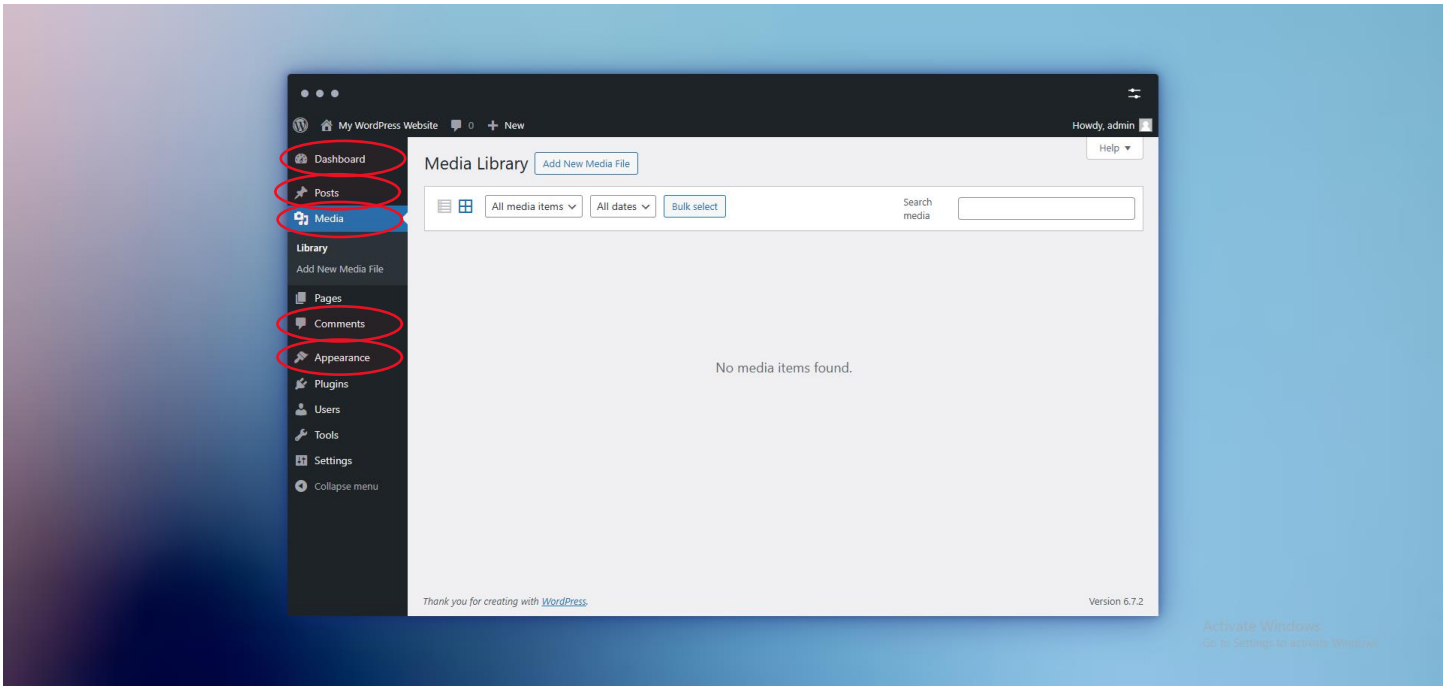
Use Case 21



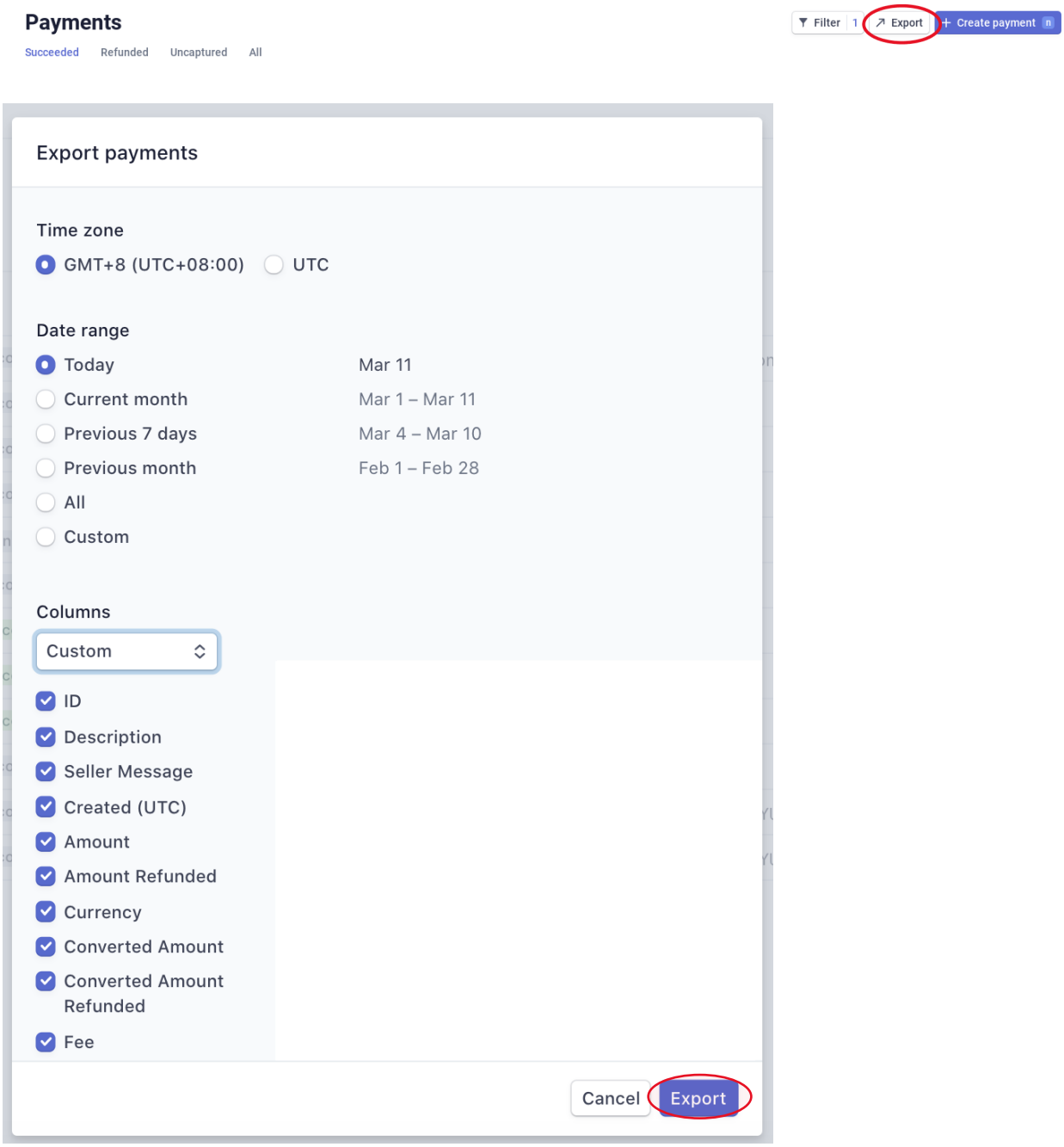
Use Case 22 - 24



Use Cases 25-29



Use Case 30


Payments
Succeeded
Refunded
Uncaptured
All
Filter
1
Export
Create payment
n
Export payments
Time zone
GMT+8 (UTC+08:00)
UTC
Date range
Today
Mar 11
Current month
Mar 1 – Mar 11
Previous 7 days
Mar 4 – Mar 10
Previous month
Feb 1 – Feb 28
All
Custom
Columns
Custom
ID
Description
Seller Message
Created (UTC)
Amount
Amount Refunded
Currency
Converted Amount
Converted Amount Refunded
Fee
Cancel
Export